

Zachary H. Jones

Performance Analyst, Computer Scientist, Software Developer,
Kernel Hacker, and Data Storyteller

EXPERIENCE

Meta — Software Engineer (and *Performance Engineer*)

MAR 2022 – PRESENT

Oversaw development and reliability of the telemetry agent, Dynolog, that runs on over 10 million servers across the fleet

- Served as tech lead for over 1 year and provided technical oversight on projects to enhance Dynolog's telemetry capabilities while navigating the team through 80% turnover, while simultaneously driving a 50% reduction in site event incidents by prioritizing data quality
- Architected the Dynolog Northstar to layout a strategic framework to navigate increasing fleet heterogeneity and complex service mixing while concurrently elevating data fidelity and system reliability across 10M+ servers.
- Developed performance modeling for key performance metrics to enable monitoring required for stacking multiple workloads per server across the fleet, driving 50% improvement in fleet efficiency
- Identified and drove reliability enhancements for Dynolog, resulting in a 50% reduction in average deployment time, an increase in uptime from 98% to 99.98%, and reduced build times
- Collaborated with hardware performance engineers to ensure Dynolog's compatibility with new hardware as it was introduced into the fleet
- As a team SME and tech leader, mentor junior engineers in performance engineering/analysis and sysops techniques

Aerospike — *Performance Engineer*

NOV 2021 – MAR 2022

Lead the performance engineering and analysis for the development of Aerospike Database Server

- Designed and implemented an automated performance testing framework to allow for quick and iterative evaluation of builds during development.
- Identified optimizations for new secondary index implementation through measuring latency & throughput and capturing flame graphs that resulted in 2-3x improvement in key workloads.
- Collaborated with Quality Engineering to identify performance regressions and with developers to mitigate regressions.

Verizon Digital Media Services (*a.k.a Edgio, Edgecast*) — *Principal Performance & Kernel Engineer*

MAR 2017 – NOV 2021

Helped charter and build a performance engineer team responsible for the server performance of 8% of the Internet.

- Improved performance by 5x across hardware specifications, kernel 10tunings, and software optimizations drove CDN capacity to reach 140Tbps.
- Led engineering efforts to improve load balancing and DDoS capacity by 2x by leveraging XDP/BPF programing.
- Devised, published, and presented a methodology for performance analysis of XDP programs.

SUMMARY

A technical leader with more than 15 years of engineering and innovation experience driving performance changes into solutions; and leading efforts across teams to wrangle, manage, and understand data.

CONTACT

me@zjones.phd

SKILLS

- Performance Engineering & Analysis
- Internet Scale System Design & Architecture
- XDP / BPF / Linux Kernel
- Data Management / Software Defined Storage
- *nix / VMs / Containers
- Python / JS / C / C++
- Jupyter / SciPy Stack

PATENTS

US 9083608 & 9088479

Automatically selecting appropriate platform to run application in cloud computing environment

US 9075643 & 9075645

Automatically selecting optimal transport protocol in a cloud computing environment

US 8965754 & 8972245

Text prediction using environment hints

US 8949848

Reducing usage of resource utilized by a virtual machine whose resource utilization is adversely affecting neighboring virtual machines

Served as Systems & Performance Architect to drive performance engineering efforts across the larger engineering and operations organizations.

- Built custom kernel infrastructure for providing performance and security enhancing patches and configuration changes.
- Steered generational server hardware specifications to right size hardware to optimally match software requirements and provide flexibility for POP expansion and new POP growth.
- Instigated AI/ML research projects to add autonomic self-protection and self-optimization features to CI/CD process to allow auto advancing and reverting of code deployments in production.
- Refined automatic provisioning workflow and debugged failures from cross-system interactions to minimize engineers time overseeing the process while maintaining high levels of customer performance.

Demonstrated leadership through advocacy and mentorship.

- Designed a performance test harness, a framework to design a performance project/infrastructure for repeatable testing, analysis, and reporting for junior engineers and interns to take ownership of and lead their projects. Promoted the use of Jupyter Notebooks to create reproducible data documents for effective sharing and collaboration of project data and improve decision making across teams.
- Provided interns with impactful and meaningful projects while providing guidance on achievable goals for their tenure.

NetApp — *Performance Analyst, Analytics Platform Developer*

DEC 2011 – MAR 2017

Technically leader for platform engineering team creating modern analytics platforms for an internal performance monitoring product and cloud-based customer monitoring product.

- Developed services to transform/query performance data generated by ONTAP and metrics engine/recipes to convert data into higher-level forms which enables decision making from developers to executives.
- Delivered an analytics platform prototype that exceeded MVP target in 8 weeks by leveraging open-source projects and AWS elastic services.

Integrated Jupyter Notebooks into the performance development and analysis workflows to allow sharing and collaboration between performance analysts, engineers, and support.

- A grassroots effort to build tools on top of the Jupyter ecosystem resulted in 100+ engineers using the tools daily for performance analysis and driving efficiency improvements.

Led analysis investigations on next-generation software defined platforms: ONTAP Select & Cloud.

- Increased performance between 100% and 300% across various platforms and workloads through strong technical partnering with engineering to ensure performance critical changes were delivered into ONTAP while keeping leadership updates on incremental improvements.

Led analysis for development of File & Free Space Reallocation defragmentation technologies in ONTAP.

- Analyzed data collected across the customer base to understand the rate of fragmentation aging on FAS platforms to guide support organizations on best practices for customers

PRIOR EXPERIENCE

IBM MAY 2007 - AUG 2009, NOV 2010 - DEC 2011

Performance Engineer Intern, Performance Engineer for WebSphere Product Family

Toyota Racing Development USA MAY 2010 - DEC 2010

Performance Engineering/HPC, Simulation Engineering, Video Processing/Data Extraction

Clemson University, School of Computing AUG 2006 - DEC 2010

Graduate Research Assistant, Sporting Events Mobile App/Infrastructure Developer, System Administrator

EDUCATION

Ph.D. Computer Science — *Clemson University* — Dec 2010

B.S. Computer Science & Mathematics — *High Point University* — May 2006

Projects, publications/presentations, and complete employment history
available on LinkedIn: <https://www.linkedin.com/in/zacharyhjones/>

